2023 - quiz2

Graded

Student

William Chen

Total Points

10 / 10 pts

Question 1

Water flows...

5 / 5 pts

✔ **– 0 pts** Correct

– 2 pts approach error

– 3 pts wrong approach

– 1 pt unit error or calculation error

Question 2

The figure...

5 / 5 pts

✔ **– 0 pts** Correct

– 2 pts approach error

– 3 pts wrong approach

– 1 pt calculation error

– 4 pts wrong method

Quiz 02

PH-UY 2023, Spring 2026
A Sections

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(1) (5 pts) Water flows through a pipe at $v_1 = 2\,\text{m/s}$ at one location and at $v_2 = 18\,\text{m/s}$ at another location. If the pipe's radius at the first location is $r_1 = 15\,\text{cm}$, what is the radius r_2 of the pipe at the second location? *(Try to be efficient about this and don't overcalculate!)*

$v_1 = 2\text{m/s}$

$v_2 = 18\text{m/s}$

$r_1 = 0.15\text{m}$ $\quad A_1 = \pi (0.15\text{m})^2 = 0.07068\text{m}^2$

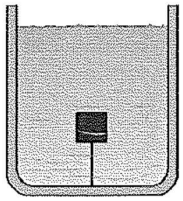
$A_1 v_1 = A_2 v_2$

$A_2 = \frac{A_1 v_1}{v_2} = \frac{1}{9} A_1 = 0.007854\text{m}^2$

$\pi r^2 = 0.007854$

$r = 0.05\text{m} = 5\text{cm}$

(2) (5 pts) The figure shows a block fully submerged in a container of water. It is attached to bottom of the container by a light rope and is in hydrostatic equilibrium. The tension in the rope is five times the weight of the block. **Determine the specific gravity of the block.** *Your answer should be a pure number!*



$F_b - F_g - T = 0$ $\quad T = 5F_g$

$F_b - 6mg = 0$

$F_b = 6mg$ $\quad F_b = 6F_g$

$F_g = \frac{1}{6} F_b$

$\frac{1}{6}$